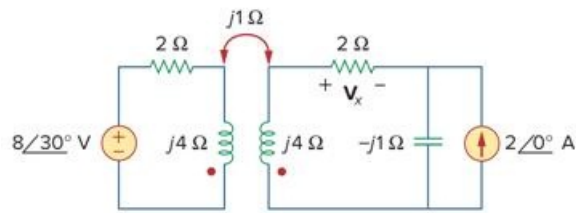


Problem

Find V_x in the network shown in Fig. 13.78.

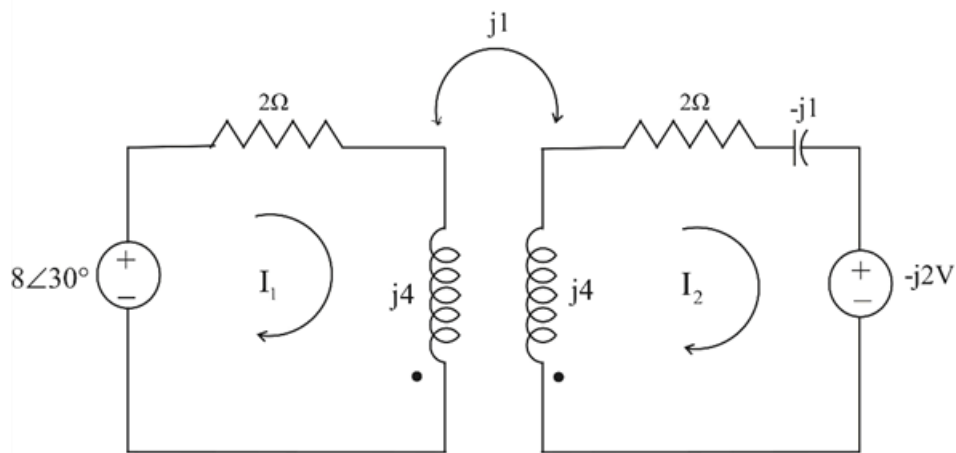
Figure 13.78



Step-by-step solution

Step 1 of 4

Consider the circuit below.



Step 2 of 4

For loop 1,

$$8\angle 30^\circ = (2 + j4)I_1 - jI_2 \rightarrow (1)$$

For loop 2,

$$(j4 + 2 - j)I_2 - jI_1 + (-j2) = 0$$

$$\text{Or, } I_1 = (3 - j2)I_2 - 2 \rightarrow (2)$$

Step 3 of 4

Substituting (2) into (1),

$$8\angle 30^\circ = (2 + j4) \left[(3 - j2)I_2 - 2 \right] - jI_2$$

$$8\angle 30^\circ = (14 + j8)I_2 - 4 - j8 - jI_2$$

$$8\angle 30^\circ = (14 + j7)I_2 - 4 - j8$$

$$I_2 = \frac{8\angle 30^\circ + 4 + j8}{14 + j7}$$

$$I_2 = \frac{6.928 + j4 + 4 + j8}{14 + j7}$$

$$I_2 = \frac{10.928 + j12}{14 + j7}$$

Step 4 of 4

$$I_2 = \frac{16.23\angle 47.68^\circ}{15.65\angle 26.56^\circ}$$

$$I_2 = 1.037\angle 21.12^\circ$$

$$V_x = 2I_2$$

$$V_x = 2(1.037\angle 21.12^\circ)$$

$$\boxed{V_x = 2.074\angle 21.12^\circ \text{ V}}$$